

Byeongjee Kang

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Research Interests

I develop programming languages and compiler techniques for safe, efficient execution across heterogeneous systems. My experience spans from energy-intermittent microcontrollers to SIMD/multicore servers. I leverage MLIR for compiler pipelines, WebAssembly for heterogeneous deployment, and Rust for developer-facing interfaces.

Education

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| PhD Carnegie Mellon University , Software Engineering
Advised by Prof. Limin Jia | Pittsburgh, USA
Aug 2023 – present |
| BS Pohang University of Science and Technology (POSTECH) , Computer Science and Engineering & Mathematics (double major) <ul style="list-style-type: none"> • GPA: 4.26/4.3 (Ranked 1st among all graduating students in class of 2023) • Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA • Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021) | Pohang, Korea
Feb 2018 – Feb 2023 |

Professional Skills

Programming Languages: Rust, C++, WebAssembly, SML, Python, JavaScript, Scala, Maude, Futhark

Frameworks: LLVM, MLIR, React, Django, Git

Work Experiences

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|---|---|
| Microsoft Research , Research intern at Data Systems Group mentored by Kaushik Rajan <ul style="list-style-type: none"> • Compiler for database intrinsics: Auto-vectorizing compiler on MLIR with support for control flow dependencies • Developed compiler optimization passes for database intrinsics using MLIR and Polygeist, improving the performance of database intrinsic functions with control dependencies through vectorization | Redmond, WA, USA
May 2025 – Aug 2025 |
| Kodebox , Software engineer as military service alternative <ul style="list-style-type: none"> • CodeChain Foundry: Open-source blockchain engine based on composable module system • Worked as core developer of consensus engine, implementing BLS signature aggregation scheme and designing interface for modules to communicate with consensus engine • Zuzu: Web-based captable management software • Worked as full-stack software engineer | Seoul, Korea
Jan 2020 – Dec 2021 |

Publications

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|---|----------|
| Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM
Ao Li, Byeongjee Kang, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye
10.1145/3764119 (Object-oriented Programming, Systems, Languages, and Applications (OOP-SLA) 2025) | Jan 2025 |
| GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits
Sam Westrick, Pengyu Liu, Byeongjee Kang, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding, Umut Acar
ieeexplore.ieee.org/document/10821338 (IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (Best Paper)) | Jan 2024 |

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems

Jan 2024

Byeongjee Kang, Kyungmin Bae

[10.1016/j.scico.2024.103097](https://doi.org/10.1016/j.scico.2024.103097) (Science of Computer Programming 2024)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search

Jan 2022

Byeongjee Kang, Kyungmin Bae

[dl.acm.org/doi/10.1145/3563822.3568017](https://doi.org/10.1145/3563822.3568017) (International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022)

Preprints

WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction

Byeongjee Kang, Harsh Desai, Limin Jia, Brandon Lucia

arxiv.org/abs/2506.16048

Mr3: An Execution Framework for Hive and Spark

Sungwoo Park, Seunggon Namgung, Byeongjee Kang

dx.doi.org/10.2139/ssrn.4587845

Talks

- An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [\[recording\]](#)

Ongoing Research Projects

Automatic Program Rescaling and Migration

May 2025 – present

Programming language and runtime support for automatically rescaling programs (including ML applications) for resource-constrained systems

- Aiming to automatically rescale programs with correctness guarantees using general approaches such as program slicing, as well as ML-specific techniques such as quantization and sparsification

Probabilistic Modeling and Analysis Tools for Intermittent System

Aug 2025 – present

Probabilistic modeling of resource usage in intermittent systems

WAMI: MLIR-based Compilation Pipeline for WebAssembly

Oct 2024 – present

MLIR-based compilation pipeline for WebAssembly

Honors and Scholarships

ILJU Fellowship (ILJU Academy and Culture Foundation)

Aug 2023 – July 2028

- Received total of USD 120,000 over eight semesters during graduate studies

Program for Highly Dedicated Students (POSTECH)

Feb 2018 – Feb 2023

- Presidential scholarship and supporting program in POSTECH offered to one outstanding freshman each year

Presidential Science Scholarship (Ministry of Education of Korea)

Feb 2018 – Feb 2023

- National science scholarship offered to top freshmen students in the field of science and technology